

abhishekh reddy munnangi

aspiring robotics software developer

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github.com/armgits

education

M.Eng in Robotics at University of Maryland, College Park

Aug 2022 - May 2024

Graduated with 3.27 GPA

B.Tech in Mechatronics Engineering at Ajeenkya DY Patil University, Pune

Aug 2018 - May 2022

Graduated with 7.87/10 CGPA

skills

Programming Languages	C++ and Python
Software Development Tools	Git, Docker, Shell Scripts, CMake and GoogleTest
Frameworks and Libraries	ROS2 (rclcpp), MoveIt!, OpenCV, Orocos-KDL, OpenGL, Eigen and NumPy
Documentation	Doxygen, Sphinx, Markdown and LaTeX

experience

Graduate Assistant at Institute for Systems Research, University of Maryland

Aug 2023 - Present

- Developing an application to track and evaluate line following robots visually using fiducial markers. Utilized modern C++ (C++11) and OOP principles, OpenCV and Matplotlib for implementation. Results serve as a rubric for grading lab exercises.
- Ported lab exercises from ROS1 to ROS2 by creating a custom Docker image and integrating it with VSCode to improve the development environment for students.
- Delivered a hotfix for a C++ wrapper library based on MoveIt! API used during the course. This significantly improved execution success rate of student-written code when using the Cartesian motion planner to move the robotic arm.
- Migrated the documentation for lab exercises from GitHub Wiki to Sphinx and hosted it on Read The Docs, resulting in centralized documentation for improved organization and maintenance.

Robotics Engineer Intern at iTrontik Smart Systems, Pune

May 2021 - Apr 2022

- Developed hardware for a nano class (less than 250 grams) quadcopter drone for aerial photography applications.
- Sized the brushless motors and propellers for the drone, selected suitable off-the-shelf parts that achieved 90 percent of the target hover flight time on average in outdoor testing, with a 12 percent excess in total weight.
- Analyzed vibrations in the frame design using flight log data and modal analysis in Fusion 360, and made improvements to the design which resulted in a 66 percent reduction of vibrations, making the drone flyable.
- Ideated and prototyped a USB-C charging solution for the drone battery pack, enabling the use of existing smartphone chargers instead of a dedicated external charger.

projects

Dijkstra Planner for Point Robot as coursework project

Mar 2024 - Apr 2024

- Implemented a Dijkstra planner in C++ on a holonomic point robot to navigate through a 2D obstacle map.
- Created a 2D visualization from scratch using OpenGL to demonstrate graph search and backtracking.
- Leveraged multi-threading to simultaneously visualize and perform the graph search.

Workspace Plotter as student club project

Jan 2024 - Feb 2024

- Developed a C++ program utilizing KDL to plot the workspace of serial manipulators using Monte Carlo simulation.
- Implemented a parser to read the DH parameters from a YAML file using yaml-cpp to rapidly test different configurations.

Mint Flight Controller Board as personal project

Oct 2020 - May 2021

- Designed a miniature flight controller board for quadcopter drones in KiCAD based on STM32H7 microcontroller.
- The resulting 4-layer PCB sizes 80 percent smaller than Pixhawk 4 while retaining most functionality.